NIRF Dashboard

Objective:
Develop an interactive dashboard using the NIRF ranking dataset covering the years 2017 to 2025 (Including all the categories like Engineering, Overall, Pharma, etc.). The dashboard should allow users to explore institutional rankings, compare institutions, identify trends, analyze geographical distributions, and investigate relationships between ranking parameters through appropriate visualizations.

Data Collection and Preparation:

1. Collect the NIRF ranking data for the years 2017 to 2025, covering all available ranking categories.
2. Perform data cleaning, preprocessing, and integration to create a consistent dataset suitable for analysis.
3. Handle changes in the number and definition of ranking categories across years. For example, NIRF covered six categories in 2017, whereas more categories were available by 2025.
4. Standardize institution names, state names, category labels, year formats, and numerical fields wherever appropriate.
5. Identify and handle missing values, duplicate records, inconsistent formats, and changes in category definitions.
6. Maintain the original category and year information wherever necessary for accurate historical comparisons.

Input and Expected Output

- Input: Analytical questions concerning any relevant aspect of the collected NIRF dataset.

- Expected output: An appropriate chart or visualization that answers the question, with meaningful titles, axis labels, legends, and filters where applicable.

- Interactive features: Users should be able to select relevant years, institutions, ranking categories, states, and parameters, depending on the question being analyzed.

Sample Questions are given in Annexure A

Project Level and Evaluation:
Level 1: Sample Questions : Implement all sample questions. Each question must produce the correct chart using the cleaned dataset. This carries **10 Marks**

Level 2: The dashboard will be tested during the final evaluation using additional questions that were not included in the sample list. Teams must demonstrate that the dashboard can dynamically generate appropriate charts using different parameters, filters, and selections. This carries **5 Marks**

Level 3: Natural-Language Query Interface : As an advanced extension, develop a chatbox that accepts questions in natural language and converts them into appropriate data queries and visualizations. This carries **5 Marks**

AI Policy:

- Teams may use any suitable programming language, library, framework, or dashboard development tool.
- AI tools may be used for coding, debugging, data cleaning, chart selection, documentation, and natural-language query implementation.
- Teams must understand their implementation and be able to explain the data-processing steps, analytical methods, and visualizations during evaluation.
- The final dashboard must run successfully and demonstrate the functionality claimed by the team.

Team Member Size: Minimum 6 and Maximum 8 .

Fill the Google form for sharing the team details (Any one of the team member can fill the form) : https://forms.gle/ymQiZi2NUjxiw7BJ9 **Deadline:** Today Evening 5 PM

Evaluation Date: November 7th 2026, Saturday - Team wise Schedule will be shared

Annexure A - Sample Questionnaire

1. Time-series and trend analysis

1. How has the overall NIRF ranking of a selected institution changed from 2017 to 2025?
2. Which institutions have shown the greatest improvement in rank over the nine years?
3. Which institutions have experienced the largest decline in ranking?
4. How has the average score of institutions changed over time?
5. Which institutions have remained consistently in the top 10 across multiple years?

2. Geospatial and regional analysis

1. Which states have the highest number of institutions in the top 100?
2. How is the geographical distribution of top-ranked institutions changing from 2017 to 2025?
3. Which states have improved their representation in the top 100 over time?
4. How do the rankings of institutions in Karnataka compare with those in Tamil Nadu?
5. Which states have produced the largest number of newly appearing institutions in the rankings?

3. Univariate analysis

1. What is the distribution of NIRF scores for a selected year?
2. What is the distribution of overall ranks across all institutions?
3. What is the average, median, minimum, and maximum score in a given year?
4. What percentage of ranked institutions fall into each score quartile?
5. How does the spread of scores differ across NIRF categories?

4. Bivariate analysis and correlation

1. How strongly are Research and Professional Practice (RPC) scores associated with overall scores?
2. How does Perception (PR) relate to overall scores?
3. Which NIRF parameter has the strongest correlation with the overall score?
4. How are TLR and RPC scores related across institutions?
5. Has the correlation between research scores and overall scores changed from 2017 to 2025?

A useful dashboard feature here would be a select-two-variables interface that allows users to choose any two numerical attributes and immediately generate a scatter plot, correlation coefficient, and optional trend line.

5. Multivariate analysis

1. How do the top five institutions compare across all five NIRF parameters?
2. Which institutions have high research scores but relatively low teaching scores?
3. Which institutions have high teaching scores but relatively low perception scores?
4. Which institutions have improved their overall scores despite declining scores in one or more parameters?
5. Which institutions have the largest difference between their strongest and weakest parameters?

6. Distribution and statistical comparison

1. How does the score distribution differ between the top 10 and the remaining institutions?
2. Are NIRF scores more dispersed in one year than another?
3. Which NIRF parameter exhibits the greatest variability across institutions?
4. How do the median and interquartile range of scores change over time?
5. How does the score distribution differ across states?

7. Ranking transitions and institutional mobility

1. How many institutions moved from the 51–100 band into the top 50 between two selected years?
2. Which institutions entered or exited the top 10 between consecutive years?
3. What percentage of institutions remained in the same rank band from one year to the next?
4. Which institutions experienced the largest upward and downward rank movements?
5. How many institutions moved up, moved down, or retained their positions each year?

8. Outlier detection and performance gaps

1. Which institutions have unusually high research scores relative to their overall scores?
2. Which institutions perform substantially better in research than in teaching?
3. Which institutions experienced an unusually large year-to-year score change?
4. Which institutions have a strong parameter score but a relatively poor overall rank?
5. Which institutions have unusually high ranking volatility compared with similar institutions?